

BEYOND VERTEX PROFILES: EDGE-LOCAL SPECTRAL INFORMATION AND SIZE-DEPENDENT SCALING IN NONUNIFORM HYPERGRAPH TENSORS

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ABSTRACT. Vertex edge-size profiles determine every slice row sum in the support-exact edge-total-mass tensor of a nonuniform hypergraph, but we prove that they do not determine its spectral radius, even when the entire labelled 2-edge layer is fixed. Among simple labelled rank-three hypergraphs with a fixed labelled 2-edge layer, five vertices are minimal for such a spectral separation. More strongly, the same profile-equivalent pair is spectrally separated for every positive choice of the 2- and 3-edge masses, so the phenomenon is not tied to a special normalization. We then identify part of the missing information. Specializing the standard one-step Collatz–Wielandt row-sum scaling to the support-exact model produces an explicit edge-local quantity satisfying $\rho(\mathcal{A}_\eta(H)) \leq B_{\text{np}}^\eta(H) \leq \max_i R_i^\eta$ that detects how profile values are placed inside individual support edges; we obtain an exact equality characterization and explicit computable defect estimates. In the uniform specialization, this edge-local bound is strictly incomparable with both classical two-largest-degree and average-2-degree upper bounds. Finally, multi-size loose stars show that lower-order edges alter the power-law growth regime itself: a dominant family of pendant s -edges has exponent $(r - s + 1)/r$, while comparable edge sizes obey an explicit balance law. Thus vertex profiles lose spectral information, edge-local placement recovers part of it, and repeated-index structure generated by edge size controls asymptotic growth.

1. INTRODUCTION

Spectral hypergraph theory extends graph spectral methods from pairwise to higher-order interactions. For uniform hypergraphs, adjacency tensors provide a standard setting for Perron–Frobenius theory, tensor eigenvalues, and spectral-radius estimates based on degree data [22, 19, 4, 20, 6, 26, 8]. For nonuniform hypergraphs, however, edges of different cardinalities must be represented in a common multilinear order, and different constructions make different choices at this step [3, 28, 24, 21, 10, 9, 1, 5]. Structural statements in the nonuniform setting must therefore be tied to the chosen tensor representation and normalization.

We work with the support-exact weighted tensor of Galuppi–Mulas–Venturello [9] under the size-homogeneous choice $w(e) = \eta_s/s$ on every s -edge. Equivalently, each s -edge carries prescribed total tensor mass η_s , distributed uniformly over the order- r words whose support is exactly that edge. The central question is not how to define another tensor model, but which combinatorial information this fixed model can and cannot recover spectrally.

Three levels of information arise naturally:

degree data $\longrightarrow$ vertex edge-size profiles $\longrightarrow$ edge-local incidence arrangement.

The edge-size profile of a vertex records the numbers $d_i^{(s)}$ of incident edges of each cardinality. Under the present normalization it determines the slice row sum

$$R_i^\eta = \sum_{s \in \mathcal{S}(H)} \frac{\eta_s}{s} d_i^{(s)}.$$

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It does not, however, record how vertices carrying prescribed profiles are grouped inside individual hyperedges. Our first main result shows that this missing arrangement has independent spectral content: two labelled, genuinely nonuniform rank-three hypergraphs can have the same full labelled collection of vertex edge-size profiles and exactly the same complete 2-edge layer, yet different spectral radii. The separation is uniform over the normalization: for the same five-vertex pair,

$$\rho(\mathcal{A}_\eta(H_1)) > \rho(\mathcal{A}_\eta(H_2)) \quad \text{for every } \eta_2, \eta_3 > 0.$$

Five vertices are minimal for this phenomenon even when the common 2-edge layer is not required to be complete. Thus the full labelled profile data, even together with all pairwise edge information, do not determine the spectral radius, and this failure persists over the entire positive cone of the two active edge-size masses.

To expose part of the missing information, we evaluate the Collatz–Wielandt quotient at the explicit profile scaling

$$z_i = (R_i^\eta)^{1/(r-1)}.$$

This yields the neighbor-profile quantity $B_{\text{np}}^\eta(H)$ and the refinement

$$\rho(\mathcal{A}_\eta(H)) \leq B_{\text{np}}^\eta(H) \leq \max_i R_i^\eta.$$

The one-step scaling mechanism itself is standard in nonnegative tensor theory [4, 26, 8, 18, 27]. Accordingly, the novelty of B_{np}^η is not the abstract Collatz–Wielandt rescaling itself. We therefore distinguish the standard analytic step from the new structural consequences. What is standard is the substitution of a positive profile-scaled test vector into the Collatz–Wielandt quotient. What is specific to the present support-exact model is the resulting edge-local closed form and the theory built from it: it detects profile-equivalent incidence arrangements, has an exact equality criterion and explicit computable defect estimates, is strictly incomparable with two classical uniform degree-based bounds, and fits the same edge-local organization used later for exact quotient reduction and asymptotic analysis. The comparison with existing spectral-radius bounds is therefore informational rather than a claim of universal numerical dominance. Classical uniform estimates use the two largest degrees or average 2-degrees [14, 13], while later work develops further degree/co-degree bounds and diagonal-scaling refinements [15, 16]. In the uniform specialization we prove that B_{np}^η is strictly incomparable with both classical baselines. Thus it is neither a uniform improvement nor a reparametrization of those aggregate statistics; its role is to retain edgewise placement information. Results for other general-hypergraph tensor settings are cited for context but are not claimed to be subsumed [7, 17].

The same edge-local organization also supports exact quotient reductions. Tensor equitable partitions are standard [11], with related hypergraph notions including perfect colourings and quotient constructions [25, 2]. We introduce an edge-type-equitable condition stated directly at the mixed-size hypergraph level and prove that it implies tensor equitability for the support-exact blowup tensor. Under weak irreducibility, this yields an exact reduction of the spectral radius. The condition is intended as a directly checkable sufficient criterion; no converse is claimed.

A third layer appears in asymptotic families. For multi-size loose stars, lower-order edges produce repeated-index monomials in the fixed-order tensor, and these repetitions change the spectral scale itself. The loose-star geometry considered here is distinct from the hyperflower family studied in [9]: a uniform loose star has a single common center and otherwise disjoint leaves, whereas a uniform hyperflower shares $r - 1$ central vertices across its edges. If pendant s -edges dominate, then

$$\rho(\mathcal{A}_\eta(H)) \asymp k_s^{(r-s+1)/r},$$

with an explicit leading constant in the one-size dominant regime. Hence edge cardinality changes the exponent, not merely a normalization factor. When several natural scales are comparable, an explicit balance equation describes their simultaneous leading-order contribution.

These results are linked by one feature of the fixed-order support-exact encoding: it preserves the identity of each support edge, while lower-order edges necessarily generate repeated indices. The same mechanism makes edgewise profile placement spectrally visible and changes the asymptotic balance between center and leaf equations.

The main contributions form three successive levels.

- (i) *Information loss.* We prove spectral non-determination beyond the full labelled collection of vertex edge-size profiles. A rank-three pair H_1, H_2 has identical labelled profiles and the same complete 2-edge layer, yet

$$\rho(\mathcal{A}_\eta(H_1)) > \rho(\mathcal{A}_\eta(H_2)) \quad \text{for every } \eta_2, \eta_3 > 0.$$

Thus the separation is uniform over all positive relative edge-size masses, not merely stable near one chosen normalization. A minimality argument shows that no rank-three profile-equivalent pair with the same labelled 2-edge layer can differ on fewer than five vertices.

- (ii) *Edge-local recovery.* Starting from the standard one-step Collatz–Wielandt scaling, we derive its explicit edge-local specialization B_{np}^η . It detects incidence arrangement beyond the labelled profiles, admits exact equality criteria and explicit computable defect estimates, and is strictly sharper than the profile row-sum baseline by an unbounded factor on an infinite structured family. Even in the uniform rank-three setting it is incomparable both with the classical Yuan–Zhang–Lu degree-sequence bound and with the Lin–Mo–Zhou–Weng average 2-degree bound. Thus the edgewise information retained by B_{np}^η is not reducible to either the two largest degrees or the corresponding aggregate average-2-degree statistics. The same edge-local organization yields a directly checkable edge-type-equitable condition implying tensor equitability and hence exact spectral-radius quotient reductions under weak irreducibility.
- (iii) *Size-dependent growth.* We determine the spectral scale of multi-size loose stars. Competing edge sizes satisfy an explicit balance law, and a dominant family of pendant s -edges has exponent $(r - s + 1)/r$. Thus edge cardinality changes the power-law growth regime itself, rather than merely a normalization constant. For r -uniform loose stars, the neighbor-profile refinement also improves the profile row-sum bound by an unbounded factor.

The paper is organized as follows. Section 2 fixes the basic tensor and hypergraph notation, while Section 3 introduces the edge-total-mass model and its vertex-profile row-sum baseline. Section 4 then follows the main spectral-information arc: robust profile non-determination, edge-local recovery through B_{np}^η , and a uniform specialization that places this quantity against classical degree and average-2-degree bounds. Section 5 isolates the equality, strictness, and defect theory of that recovery step. Section 6 turns the same edge-local organization into exact quotient reductions, which are then used in Section 7 to derive the size-dependent loose-star scaling laws.

2. PRELIMINARIES

2.1. Hypergraphs and edge-size profiles. Let $H = (V, E)$ be a finite simple hypergraph with at least one edge, with vertex set $V = \{1, \dots, n\}$, and with edge set $E \subseteq 2^V$. Here “simple” means that edges are not repeated; containment between distinct edges is allowed, so no Sperner condition is imposed. Throughout the paper, every edge is assumed to have cardinality at least 2. The rank of H is

$$r = r(H) = \max_{e \in E} |e|.$$

We assume throughout that $r \geq 3$, so that at least one edge has cardinality greater than two. The case $r = 2$ reduces to the matrix setting and is excluded from the main statements.

For $2 \leq s \leq r$, set

$$E_s = \{e \in E : |e| = s\}.$$

The active edge-size set of H is

$$\mathcal{S}(H) = \{s \in \{2, \dots, r\} : E_s \neq \emptyset\}.$$

This set is nonempty because H has at least one edge. Edge sizes not belonging to $\mathcal{S}(H)$ do not occur in the hypergraph and produce no tensor entries or profile contributions. We therefore index all edge-size weights only by $\mathcal{S}(H)$, rather than by dormant sizes that are absent from H .

For each vertex $i \in V$, define the s -degree

$$d_i^{(s)} = |\{e \in E_s : i \in e\}|.$$

The vector

$$\mathbf{d}_i = (d_i^{(2)}, d_i^{(3)}, \dots, d_i^{(r)})$$

is the vertex edge-size profile of i , with $d_i^{(s)} = 0$ whenever $s \notin \mathcal{S}(H)$. The ordinary degree is

$$d_i = \sum_{s=2}^r d_i^{(s)} = \sum_{s \in \mathcal{S}(H)} d_i^{(s)}.$$

We denote the maximum degree by

$$\Delta = \max_i d_i.$$

Definition 2.1 (Profile-regularity). The hypergraph H is called profile-regular if

$$\mathbf{d}_1 = \mathbf{d}_2 = \dots = \mathbf{d}_n.$$

After a weight vector η is fixed, H is called η -row-regular if its profile row sums satisfy

$$R_1^\eta = R_2^\eta = \dots = R_n^\eta.$$

Definition 2.2 (Profile equivalence). Two labelled hypergraphs H and H' on the same vertex set and with the same rank are called profile-equivalent if

$$d_i^{(s)}(H) = d_i^{(s)}(H')$$

for every vertex i and every size $2 \leq s \leq r$.

Isolated vertices have all profile components equal to zero. They produce zero slices in the adjacency tensor and do not affect the spectral radius. Let

$$V^+ = \{i \in V : d_i > 0\}$$

be the non-isolated support of H , and let H^+ be the hypergraph obtained by deleting isolated vertices. Since isolated vertices generate zero tensor slices and do not occur in any edge, the tensor $\mathcal{A}_\eta(H)$ decomposes as $\mathcal{A}_\eta(H^+)$ together with zero slices. Hence

$$\rho(\mathcal{A}_\eta(H)) = \rho(\mathcal{A}_\eta(H^+)).$$

Accordingly, the neighbor-profile quantities below are stated for hypergraphs without isolated vertices, or equivalently applied to H^+ if isolated vertices are present in the original hypergraph.

2.2. Standing assumptions and exceptions. Unless stated otherwise, H is finite and simple, every edge has size at least two, $r \geq 3$, and $\eta_s > 0$ for every active size $s \in \mathcal{S}(H)$. Isolated vertices may be deleted without changing the spectral radius, and all neighbor-profile formulas are applied to the non-isolated support. Connectivity is not required for the row-sum, Collatz–Wielandt, neighbor-profile, or defect bounds. Weak irreducibility is used only when a strictly positive Perron vector or an exact quotient reduction is required; for $\mathcal{A}_\eta(H)$ it is equivalent to connectivity of the non-isolated 2-section.

2.3. Tensor eigenvalues and Collatz–Wielandt bounds. Let $\mathcal{A} = (a_{i_1 \dots i_r})$ be a real nonnegative tensor of order r and dimension n . For $x \in \mathbb{C}^n$, define

$$(\mathcal{A}x^{r-1})_i = \sum_{i_2, \dots, i_r=1}^n a_{ii_2 \dots i_r} x_{i_2} \cdots x_{i_r}.$$

When $\mathcal{A}$ is symmetric, we also write

$$\mathcal{A}x^r = \sum_{i_1, \dots, i_r=1}^n a_{i_1 \dots i_r} x_{i_1} \cdots x_{i_r}.$$

With this convention,

$$\frac{\partial}{\partial x_i} (\mathcal{A}x^r) = r(\mathcal{A}x^{r-1})_i.$$

A tensor eigenpair is a pair $(\lambda, x) \in \mathbb{C} \times (\mathbb{C}^n \setminus \{0\})$ such that

$$\mathcal{A}x^{r-1} = \lambda x^{[r-1]}, \quad x^{[r-1]} = (x_1^{r-1}, \dots, x_n^{r-1}).$$

The spectral radius $\rho(\mathcal{A})$ is the maximum modulus of all tensor eigenvalues. When x is real and λ is real, the pair is an H -eigenpair. For nonnegative tensors, Perron–Frobenius theory ensures that $\rho(\mathcal{A})$ is an eigenvalue with a nonnegative eigenvector.

For a symmetric nonnegative tensor we also use the standard variational characterization

$$(2.1) \quad \rho(\mathcal{A}) = \max \left\{ \mathcal{A}x^r : x \in \mathbb{R}_+^n, \sum_{i=1}^n x_i^r = 1 \right\};$$

see, for example, Qi [23, Theorems 1–2]. In particular, every nonnegative test vector gives a lower bound through its homogeneous Rayleigh quotient.

We use the following standard Collatz–Wielandt type estimates for nonnegative tensors; see, for example, [4, 26, 8]. For every vector $x \in \mathbb{R}_{++}^n$,

$$(2.2) \quad \rho(\mathcal{A}) \leq \max_{1 \leq i \leq n} \frac{(\mathcal{A}x^{r-1})_i}{x_i^{r-1}},$$

and

$$(2.3) \quad \min_{1 \leq i \leq n} \frac{(\mathcal{A}x^{r-1})_i}{x_i^{r-1}} \leq \rho(\mathcal{A}).$$

The upper estimate is the main tool used for the general bounds. The lower estimate is recorded for completeness; the strengthened spectral-separation theorem in Section 4 instead uses the symmetric-tensor variational characterization above. These inequalities do not require irreducibility.

For reference, let $G(\mathcal{A})$ denote the representation digraph of $\mathcal{A}$: there is an arc from i to j if some positive entry $a_{ii_2 \dots i_r}$ has $j \in \{i_2, \dots, i_r\}$. Equivalently, $G(\mathcal{A})$ is the digraph of the nonnegative representation matrix whose (i, j) entry sums the entries in the i th tensor slice whose tail indices contain j . A nonnegative tensor is weakly irreducible precisely when this digraph is strongly connected. For a symmetric nonnegative tensor, this is equivalent to connectivity of the underlying undirected graph of $G(\mathcal{A})$. If $\mathcal{A}$ is weakly irreducible and nonnegative, then its positive Perron eigenvector is unique up to multiplication by a positive scalar [8, 23].

For the tensor $\mathcal{A}_\eta(H)$ defined below, the underlying graph of $G(\mathcal{A}_\eta(H))$ is the 2-section of H , namely the graph on V in which two distinct vertices are adjacent whenever they lie in a common hyperedge. Thus the 2-section is connected exactly when any two distinct vertices can be linked by a chain of overlapping hyperedges. Hence $\mathcal{A}_\eta(H)$ is weakly irreducible whenever the 2-section of H is connected. If H has several nontrivial connected components, all bounds in this paper apply componentwise, and the spectral radius is the maximum of the spectral radii of the component tensors.

3. SUPPORT-EXACT TENSOR AND VERTEX-PROFILE BASELINE

3.1. Edge-total-mass tensor. We now record the size-homogeneous specialization described in the Introduction in the notation used throughout the paper. Let $e \subseteq V$ be an edge with $|e| = s \leq r$. Define its ordered blowup set by

$$\beta(e) = \{(i_1, \dots, i_r) \in e^r : \text{supp}(i_1, \dots, i_r) = e\}.$$

Here $\text{supp}(i_1, \dots, i_r)$ denotes the set of distinct vertices appearing among the coordinates. Equivalently, every vertex of e appears at least once among $i_1, \dots, i_r$. Its cardinality is

$$(3.1) \quad |\beta(e)| = |e|! S(r, |e|),$$

where $S(r, |e|)$ is a Stirling number of the second kind. Since $|e| \leq r$, this number is positive.

Fix a positive edge-size weight vector

$$\eta = (\eta_s)_{s \in \mathcal{S}(H)}, \quad \eta_s > 0.$$

The edge-total-mass normalized tensor $\mathcal{A}_\eta(H)$ is the order- r , dimension- n symmetric tensor defined by

$$(3.2) \quad (\mathcal{A}_\eta(H))_{i_1 \dots i_r} = \begin{cases} \frac{\eta_{|e|}}{|\beta(e)|} = \frac{\eta_{|e|}}{|e|!S(r, |e|)}, & \text{if } (i_1, \dots, i_r) \in \beta(e) \text{ for some } e \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Simplicity of H ensures that a support set identifies at most one edge. Thus the definition is unambiguous. An edge e of size $|e|$ has total tensor mass $\eta_{|e|}$, distributed equally over its $|\beta(e)| = |e|!S(r, |e|)$ valid ordered blowup entries.

Remark 3.1 (Repetitions inside lower-order edges). When $|e| < r$, entries generated by e necessarily contain repeated vertices. For example, if $r = 3$ and $e = \{i, j\}$, then the valid ordered triples are precisely those over $\{i, j\}$ in which both i and j occur. This is an intentional feature of the fixed-order blowup model, not a claim that a lower-order edge has become a simple graph edge inside the tensor. The normalization below tracks the total mass assigned to this blown-up edge.

Remark 3.2 (Normalization comparison). Galuppi–Mulas–Venturello [9] assign an s -edge of weight $w(e)$ the entry weight $w(e)/((s-1)!S(r, s))$ on each valid order- r word with support e . Hence (3.2) is their tensor under $w(e) = \eta_s/s$. The unweighted Banerjee–Char–Mondal tensor [3] corresponds to $\eta_s = s$, and in the r -uniform case $\eta_r = r$ recovers the Cooper–Dutle entry weight $1/(r-1)!$.

3.2. Profile row sums. For a nonnegative tensor $\mathcal{A}$, define the slice row sum at vertex i by

$$R_i(\mathcal{A}) = \sum_{i_2, \dots, i_r=1}^n a_{ii_2 \dots i_r}.$$

For the tensor $\mathcal{A}_\eta(H)$, we write

$$R_i^\eta = R_i(\mathcal{A}_\eta(H)).$$

Under $w(e) = \eta_{|e|}/|e|$, the weighted-degree identity of Galuppi–Mulas–Venturello [9] specializes to

$$R_i^\eta = \sum_{s \in S(H)} \frac{\eta_s}{s} d_i^{(s)}.$$

We record the short support-counting proof only to fix notation and make the edge-size dependence explicit.

For $2 \leq s \leq r$, let

$$(3.3) \quad N_{r,s} = (s-1)!S(r, s) = \frac{s!S(r, s)}{s}.$$

This is the number of tail sequences $(i_2, \dots, i_r)$ that complete a fixed first vertex $i \in e$, $|e| = s$, to an order- r word with support exactly e . Indeed, the $s!S(r, s)$ surjective words onto e split equally according to their first coordinate.

Lemma 3.3 (Single-edge row contribution). *Let $e \in E$ and $i \in e$. The contribution of e to the i th row sum of $\mathcal{A}_\eta(H)$ is $\eta_{|e|}/|e|$. If $i \notin e$, the contribution is 0.*

Proof. Fix $e \in E$ with $|e| = s$, and let $i \in e$. We count the ordered tuples $(i_2, \dots, i_r)$ such that

$$\text{supp}(i, i_2, \dots, i_r) = e.$$

The number of such tuples is $N_{r,s} = s!S(r, s)/s$ by (3.3). Each valid tensor entry has weight $\eta_s/(s!S(r, s))$. Hence the total contribution is

$$\frac{s!S(r, s)}{s} \cdot \frac{\eta_s}{s!S(r, s)} = \frac{\eta_s}{s} = \frac{\eta_{|e|}}{|e|}.$$

If $i \notin e$, no tuple in the i th slice has support equal to e , so the contribution is 0. $\square$

Proposition 3.4 (Profile row-sum identity). *For every vertex $i \in V$,*

$$(3.4) \quad R_i^\eta = \sum_{s \in \mathcal{S}(H)} \frac{\eta_s}{s} d_i^{(s)}.$$

Proof. By Lemma 3.3, every incident edge e of size $|e|$ contributes $\eta_{|e|}/|e|$ to R_i^η , and every nonincident edge contributes 0. Grouping incident edges by their sizes gives (3.4). $\square$

Corollary 3.5 (Profile row-sum bound). *The spectral radius of $\mathcal{A}_\eta(H)$ satisfies*

$$(3.5) \quad \rho(\mathcal{A}_\eta(H)) \leq \max_{i \in V} R_i^\eta = \max_{i \in V} \sum_{s \in \mathcal{S}(H)} \frac{\eta_s}{s} d_i^{(s)}.$$

Proof. Apply (2.2) with $x = \mathbf{1}$. Then

$$(\mathcal{A}_\eta(H) \mathbf{1}^{r-1})_i = R_i^\eta,$$

and the claim follows from Proposition 3.4. $\square$

4. FROM PROFILE NON-DETERMINATION TO EDGE-LOCAL RECOVERY

The profile row sums retain only vertex-local information. We first show that this information is insufficient even when the entire pairwise edge layer is held fixed. We then identify an explicit edge-local refinement that detects part of the missing incidence arrangement.

4.1. Spectral non-determination and minimality. We first isolate the non-determination phenomenon before introducing any new spectral bound. The complete 2-edge layer is fixed, so any separation must come from the organization of the higher-order edges. In fact, the same five-vertex pair is separated for every positive choice of the two active edge-size masses, not merely near a selected normalization.

Theorem 4.1 (Spectral non-determination for all positive masses). *There exist genuinely nonuniform rank-three hypergraphs with identical labelled edge-size profiles, the same complete 2-edge layer (hence the same complete 2-section), and identical profile row-sum bounds, but with different spectral radii for every positive choice of the active edge-size masses.*

More explicitly, let $V = \{1, 2, 3, 4, 5\}$, let both hypergraphs contain every 2-edge,

$$E_2(H_1) = E_2(H_2) = \binom{V}{2},$$

and let their 3-edge layers be

$$E_3(H_1) = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{3, 4, 5\}\},$$

$$E_3(H_2) = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 5\}, \{2, 4, 5\}\}.$$

Then for every $\eta_2, \eta_3 > 0$ the two hypergraphs are profile-equivalent and satisfy

$$R_1^\eta = R_2^\eta = 2\eta_2 + \eta_3, \quad R_3^\eta = R_4^\eta = R_5^\eta = 2\eta_2 + \frac{2}{3}\eta_3,$$

so their profile row-sum bounds agree, while

$$\rho(\mathcal{A}_\eta(H_1)) > \rho(\mathcal{A}_\eta(H_2)).$$

Proof. The profile and row-sum statements follow exactly as in Proposition 3.4: every vertex lies in four 2-edges, vertices 1, 2 have 3-degree three, and vertices 3, 4, 5 have 3-degree two.

It remains to compare the spectral radii. Since the common 2-edge layer is complete, both tensors are weakly irreducible. Let

$$x = (x_1, \dots, x_5) \in \mathbb{R}_{++}^5, \quad \sum_{i=1}^5 x_i^3 = 1,$$

be the normalized Perron vector of $\mathcal{A}_\eta(H_2)$. The hypergraph H_2 is invariant under the permutation

$$\sigma = (1 \ 2)(3 \ 4),$$

which interchanges the 3-edges $\{1, 2, 3\}$ and $\{1, 2, 4\}$, and also interchanges $\{1, 3, 5\}$ and $\{2, 4, 5\}$. Therefore σx is another positive Perron vector with the same normalization. Uniqueness of the normalized positive Perron vector therefore gives

$$x_1 = x_2 =: a, \quad x_3 = x_4 =: b, \quad x_5 =: c.$$

Moreover $a \neq b$. Indeed, if $a = b$, then $x = (a, a, a, a, c)$. For vertices 1 and 3, the complete 2-edge layer sees the same source value a and the same multiset of the other coordinate values, namely three copies of a and one copy of c ; hence its contributions to the two tensor equations are equal. By contrast, the 3-edge contributions at vertices 1 and 3 are respectively

$$\frac{\eta_3}{3}(2ab + bc) \quad \text{and} \quad \frac{\eta_3}{3}(a^2 + ac).$$

If $a = b$, their difference is $\eta_3 a^2/3 > 0$. This contradicts the Perron equations, whose left-hand sides at vertices 1 and 3 would both equal $\rho(\mathcal{A}_\eta(H_2))a^2$.

The two tensors have the same 2-edge layer and share the 3-edges $\{1, 2, 3\}$ and $\{1, 2, 4\}$. Under the present normalization, a rank-three 3-edge $\{i, j, k\}$ contributes exactly $\eta_3 x_i x_j x_k$ to the homogeneous form, because its six ordered entries each have weight $\eta_3/6$. Consequently their homogeneous forms differ only on the remaining two 3-edges, and

$$\begin{aligned} \mathcal{A}_\eta(H_1)x^3 - \mathcal{A}_\eta(H_2)x^3 &= \eta_3 x_5(x_1 x_2 + x_3 x_4 - x_1 x_3 - x_2 x_4) \\ &= \eta_3 x_5(x_1 - x_4)(x_2 - x_3) \\ &= \eta_3 c(a - b)^2 > 0. \end{aligned}$$

Since x is the normalized Perron vector of $\mathcal{A}_\eta(H_2)$,

$$\mathcal{A}_\eta(H_2)x^3 = \rho(\mathcal{A}_\eta(H_2)).$$

The variational characterization (2.1) now yields

$$\rho(\mathcal{A}_\eta(H_1)) \geq \mathcal{A}_\eta(H_1)x^3 > \mathcal{A}_\eta(H_2)x^3 = \rho(\mathcal{A}_\eta(H_2)),$$

which proves the claim for every $\eta_2, \eta_3 > 0$. $\square$

Remark 4.2 (Uniformity over the positive mass cone). Theorem 4.1 is stronger than a continuity-based robustness statement: the same labelled pair is separated for every positive ratio η_3/η_2 . The factorization in the proof identifies the mechanism directly. At the Perron vector of H_2 , the two noncommon 3-edge arrangements differ by the positive quantity $\eta_3 c(a - b)^2$.

Proposition 4.3 (Minimality of the five-vertex separation). *Five vertices are minimal for a nontrivial rank-three profile-equivalent pair with the same labelled 2-edge layer. More precisely, let H and H' be rank-three hypergraphs on the same labelled vertex set V , with $|V| \leq 4$, identical 2-edge layers, and identical labelled edge-size profiles. Then $H = H'$.*

Proof. For $|V| = 3$, rank three forces the unique 3-edge V to be present, so the claim is immediate. Smaller vertex sets cannot have rank three. It remains to consider $|V| = 4$.

Write $e_j = V \setminus \{j\}$ and let $x_j \in \{0, 1\}$ indicate whether e_j belongs to the 3-edge layer of H . Put $m = \sum_{j \in V} x_j$. Since e_j contains vertex i exactly when $j \neq i$,

$$d_i^{(3)} = \sum_{j \neq i} x_j = m - x_i.$$

Also $\sum_i d_i^{(3)} = 3m$, so the labelled 3-degree vector determines

$$m = \frac{1}{3} \sum_i d_i^{(3)}, \quad x_i = m - d_i^{(3)}.$$

Hence the labelled 3-degree vector uniquely determines every 3-edge indicator x_i and therefore the entire 3-edge layer. Because the 2-edge layers of H and H' are assumed identical, profile equivalence forces $H = H'$. The five-vertex pair in Theorem 4.1 is therefore minimal even without requiring the common 2-edge layer to be complete. $\square$

Remark 4.4 (Role of the complete 2-edge layer). The complete 2-edge layer in Theorem 4.1 is imposed to control pairwise information as strongly as possible: the certified separation cannot be attributed to any change in pairwise adjacency. Proposition 4.3 is stronger than this constructional choice—its minimality conclusion requires only that the two 2-edge layers agree, not that they be complete.

4.2. Neighbor-profile refinement. The separation above shows that labelled vertex profiles do not contain all spectral information. We now expose part of the missing information by incorporating the profile row sums of vertices that lie in common edges. Since R_j^η is the weighted scalar image of the full edge-size profile of vertex j , the resulting quantity is called a neighbor-profile bound. The term *neighbor-profile* refers to the edgewise arrangement of the profile-induced row sums R_j^η , rather than to the full profile vector itself. Its additional content comes from how these scalar profile images are placed together inside individual hyperedges.

For an edge $e \in E$ and a vertex $i \in e$, define the edge polynomial

$$(4.1) \quad P_{r,e,i}(z) = \sum_{\substack{(i_2, \dots, i_r) \in e^{r-1} \\ \text{supp}(i, i_2, \dots, i_r) = e}} z_{i_2} \cdots z_{i_r}.$$

This is a homogeneous polynomial of degree $r - 1$ in the variables indexed by e . It records all valid repetitions of the vertices of e in the order- r tensor slice with first index i . The number of monomials in $P_{r,e,i}$, counted with multiplicity, is $N_{r,|e|}$.

For later use, define

$$\mathcal{T}(e, i) = \{(i_2, \dots, i_r) \in e^{r-1} : \text{supp}(i, i_2, \dots, i_r) = e\}.$$

Thus

$$P_{r,e,i}(z) = \sum_{(i_2, \dots, i_r) \in \mathcal{T}(e, i)} z_{i_2} \cdots z_{i_r}.$$

Assume in this section that H has no isolated vertices, so that $R_i^\eta > 0$ for all i . Define

$$z_i = (R_i^\eta)^{1/(r-1)}.$$

Theorem 4.5 (Neighbor-profile bound). *Let H be a finite simple hypergraph without isolated vertices, and let $\mathcal{A}_\eta(H)$ be the edge-total-mass normalized tensor defined in (3.2). Then*

$$(4.2) \quad \rho(\mathcal{A}_\eta(H)) \leq B_{\text{np}}^\eta(H),$$

where

$$(4.3) \quad B_{\text{np}}^\eta(H) = \max_{i \in V} \frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} P_{r,e,i}((R^\eta)^{1/(r-1)}).$$

Here $(R^\eta)^{1/(r-1)}$ denotes the vector whose j th coordinate is $(R_j^\eta)^{1/(r-1)}$. Moreover,

$$(4.4) \quad B_{\text{np}}^\eta(H) \leq \max_i R_i^\eta.$$

Proof. Since H has no isolated vertices, $R_i^\eta > 0$ for every i , and therefore $z = (z_i)$ is a strictly positive vector. Apply the Collatz–Wielandt upper bound (2.2) with $z_i = (R_i^\eta)^{1/(r-1)}$. Since $z_i^{r-1} = R_i^\eta$, we have

$$\frac{(\mathcal{A}_\eta(H) z^{r-1})_i}{z_i^{r-1}} = \frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} P_{r,e,i}((R^\eta)^{1/(r-1)}).$$

Taking the maximum over i gives (4.2).

Let $M = \max_i R_i^\eta$. Since

$$z_j^{r-1} = R_j^\eta \leq M$$

for every j , we have $z_j \leq M^{1/(r-1)}$. Therefore every monomial

$$z_{i_2} \cdots z_{i_r}$$

appearing in $P_{r,e,i}(z)$ is bounded above by

$$(M^{1/(r-1)})^{r-1} = M.$$

Since $P_{r,e,i}$ contains $N_{r,|e|}$ monomials counted with multiplicity,

$$P_{r,e,i}(z) \leq MN_{r,|e|}.$$

Therefore

$$\begin{aligned} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|!S(r, |e|)} P_{r,e,i}(z) &\leq \sum_{e \ni i} \frac{\eta_{|e|}}{|e|!S(r, |e|)} MN_{r,|e|} \\ &= M \sum_{e \ni i} \frac{\eta_{|e|}}{|e|} = MR_i^\eta. \end{aligned}$$

Dividing by R_i^η and taking the maximum over i proves (4.4). $\square$

Corollary 4.6 (Neighbor-profile separation of the minimal pair). *For the hypergraphs H_1, H_2 in Theorem 4.1, take the equal-mass normalization $\eta_2 = \eta_3 = 1$. Then*

$$B_{\text{np}}^\eta(H_1) = \frac{17}{18} + \frac{4\sqrt{2}}{3}, \quad B_{\text{np}}^\eta(H_2) = \frac{67}{54} + \frac{10\sqrt{2}}{9},$$

and therefore $B_{\text{np}}^\eta(H_1) > B_{\text{np}}^\eta(H_2)$. Thus the same edge-local arrangement that changes the true spectral radius is already detected by the explicit one-step profile scaling.

Proof. For the neighbor-profile bound, put

$$a = 3, \quad b = \frac{8}{3}, \quad s = \sqrt{ab} = 2\sqrt{2}.$$

Thus the profile test vector has value $\sqrt{a}$ on vertices 1, 2 and $\sqrt{b}$ on vertices 3, 4, 5. For a 2-edge $e = \{i, j\}$,

$$P_{3,e,i}(z) = 2z_i z_j + z_j^2.$$

Direct substitution into (4.3) gives, for H_1 , the two quotient values

$$Q_{1,h} = \frac{17}{18} + \frac{4\sqrt{2}}{3}, \quad Q_{1,l} = \frac{25}{12} + \frac{\sqrt{2}}{2}.$$

Since

$$Q_{1,h} - Q_{1,l} = \frac{-41 + 30\sqrt{2}}{36} > 0,$$

we obtain $B_{\text{np}}^\eta(H_1) = Q_{1,h}$. For H_2 , the three distinct quotient values are

$$Q_{2,h} = \frac{67}{54} + \frac{10\sqrt{2}}{9}, \quad Q_{2,m} = \frac{7}{4} + \frac{3\sqrt{2}}{4}, \quad Q_{2,5} = \frac{11}{8} + \sqrt{2}.$$

Moreover,

$$Q_{2,h} - Q_{2,m} = \frac{-55 + 39\sqrt{2}}{108} > 0, \quad Q_{2,h} - Q_{2,5} = \frac{-29 + 24\sqrt{2}}{216} > 0.$$

Hence $B_{\text{np}}^\eta(H_2) = Q_{2,h}$, and

$$B_{\text{np}}^\eta(H_1) - B_{\text{np}}^\eta(H_2) = \frac{2(3\sqrt{2} - 4)}{27} > 0.$$

$\square$

TABLE 1. At $\eta_2 = \eta_3 = 1$, the minimal pair has identical vertex-local and pairwise data, while both the edge-local refinement and the spectral radius distinguish the two higher-order incidence arrangements.

	H_1	H_2
Complete 2-edge layer	K_5	K_5
Labelled $(d_i^{(2)}, d_i^{(3)})$	$(4, 3), (4, 3), (4, 2), (4, 2), (4, 2)$	$(4, 3), (4, 3), (4, 2), (4, 2), (4, 2)$
$\max_i R_i^\eta$	3	3
B_{np}^η	$\frac{17}{18} + \frac{4\sqrt{2}}{3}$	$\frac{67}{54} + \frac{10\sqrt{2}}{9}$
Spectral ordering	$\rho(\mathcal{A}_\eta(H_1))$	$> \rho(\mathcal{A}_\eta(H_2))$

Remark 4.7 (Structural role of the one-step profile scaling). The test vector $z_i = (R_i^\eta)^{1/(r-1)}$ is a standard one-step row-sum rescaling; iterative diagonal-similarity schemes may continue such rescalings toward the spectral radius [27]. We stop after one explicit step because the contribution here is structural rather than algorithmic. The resulting $B_{\text{np}}^\eta(H)$ has an explicit edge-local form and, at equal mass, already distinguishes the minimal pair of Theorem 4.1, whose labelled profiles, row sums, edge masses, and complete 2-edge layer all agree. This model-specific separation shows that the extra information lies in the higher-order incidence arrangement, not in pairwise adjacency.

4.3. Uniform specialization and comparison with classical bounds. Suppose H is r -uniform, where $r \geq 3$. Then $\mathcal{S}(H) = \{r\}$, $d_i^{(r)} = d_i$, and $d_i^{(s)} = 0$ for $s < r$. Hence

$$(4.5) \quad R_i^\eta = \frac{\eta_r}{r} d_i.$$

The profile row-sum bound therefore reduces to the ordinary degree-type bound scaled by the normalization factor η_r/r :

$$\rho(\mathcal{A}_\eta(H)) \leq \frac{\eta_r}{r} \Delta.$$

In particular, choosing $\eta_r = r$ gives $R_i^\eta = d_i$, which matches the standard uniform adjacency tensor normalization. Thus, in the uniform case, the vertex edge-size profile contains no information beyond the ordinary degree. The point of the profile construction is its ability to distinguish genuinely nonuniform local edge-size distributions.

For the neighbor-profile bound in the r -uniform case, every valid tail tuple for an edge $e \ni i$ is a permutation of the $r - 1$ vertices in $e \setminus \{i\}$. Thus

$$P_{r,e,i}((R^\eta)^{1/(r-1)}) = (r-1)! \prod_{j \in e \setminus \{i\}} (R_j^\eta)^{1/(r-1)}.$$

Since $r!S(r, r) = r!$, the neighbor-profile bound becomes

$$B_{\text{np}}^\eta(H) = \max_i \frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_r}{r} \prod_{j \in e \setminus \{i\}} (R_j^\eta)^{1/(r-1)}.$$

Writing $a = \eta_r/r$ and using $R_i^\eta = ad_i$, this is equivalently

$$B_{\text{np}}^\eta(H) = a \max_i \frac{1}{d_i} \sum_{e \ni i} \left(\prod_{j \in e \setminus \{i\}} d_j \right)^{1/(r-1)}.$$

This differs from the average 2-degree statistic used in uniform-hypergraph spectral bounds [13],

$$m_i = \frac{\sum_{e \ni i} \prod_{j \in e \setminus \{i\}} d_j}{d_i^{r-1}}.$$

The two statistics are nevertheless linked locally by a simple power-mean inequality.

Lemma 4.8 (Local bridge to average 2-degree). *For each vertex i of an r -uniform hypergraph without isolated vertices, let*

$$\beta_i = \frac{1}{d_i} \sum_{e \ni i} \left(\prod_{j \in e \setminus \{i\}} d_j \right)^{1/(r-1)},$$

so that $B_{\text{np}}^\eta = a \max_i \beta_i$. Then

$$\beta_i^{r-1} \leq d_i^{r-2} m_i, \quad a \beta_i \leq a (d_i^{r-2} m_i)^{1/(r-1)}.$$

Proof. Apply the power-mean inequality to the d_i positive numbers

$$q_e = \left(\prod_{j \in e \setminus \{i\}} d_j \right)^{1/(r-1)}, \quad e \ni i.$$

Then

$$\left(\frac{1}{d_i} \sum_{e \ni i} q_e \right)^{r-1} \leq \frac{1}{d_i} \sum_{e \ni i} q_e^{r-1} = \frac{1}{d_i} \sum_{e \ni i} \prod_{j \in e \setminus \{i\}} d_j = d_i^{r-2} m_i.$$

Taking $r - 1$ st roots and then multiplying by $a > 0$ gives the second form. $\square$

Thus the neighbor-profile quantity does not collapse to the standard average 2-degree: it takes an edgewise geometric mean before aggregating over the incident edges. Lemma 4.8 gives a pointwise bridge, but it does not impose a global ordering between the corresponding spectral upper bounds. The distinction is not merely formal. Even within connected 3-uniform hypergraphs, the neighbor-profile bound is incomparable both with the classical degree-sequence bound of Yuan–Zhang–Lu [14] and with the average 2-degree bound of Lin–Mo–Zhou–Weng [13]. The next two propositions make both comparisons explicit. Recent work of Deng–Dong–Li further develops uniform-hypergraph estimates through degrees, average 2-degrees, and diagonal scalings [16]; our statements below do not assert a universal ordering with those later refinements or with co-degree-based estimates [15].

Proposition 4.9 (Incomparability with a classical degree-sequence bound). *Let H be a connected r -uniform hypergraph, let*

$$a = \frac{\eta_r}{r},$$

and write its degrees in nonincreasing order as $d_{(1)} \geq d_{(2)} \geq \dots$. Under the present normalization, the classical degree-sequence estimate of Yuan–Zhang–Lu gives

$$\rho(\mathcal{A}_\eta(H)) \leq U_{\text{deg}}^\eta(H) := a d_{(1)}^{1/r} d_{(2)}^{1-1/r}.$$

Neither $B_{\text{np}}^\eta(H) \leq U_{\text{deg}}^\eta(H)$ nor $U_{\text{deg}}^\eta(H) \leq B_{\text{np}}^\eta(H)$ holds for all connected uniform hypergraphs. This incomparability already occurs for $r = 3$.

Proof. For the first direction, let G_1 be the 3-uniform hypergraph on $\{1, 2, 3, 4, 5\}$ with edge set

$$E(G_1) = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 2, 5\}, \{3, 4, 5\}\}.$$

Its degree sequence is $(3, 3, 2, 2, 2)$, so

$$U_{\text{deg}}^\eta(G_1) = 3a.$$

From the uniform formula for B_{np}^η , the quotients at vertices 1, 2 are

$$a\sqrt{6},$$

whereas the quotients at vertices 3, 4, 5 are

$$\frac{a}{2}(3 + 2) = \frac{5a}{2}.$$

Since $\sqrt{6} < 5/2$,

$$B_{\text{np}}^\eta(G_1) = \frac{5a}{2} < 3a = U_{\text{deg}}^\eta(G_1).$$

For the reverse direction, let $G_2 = S_{3,k}$ be the 3-uniform loose star with $k \geq 2$ edges. Its two largest degrees are k and 1, hence

$$U_{\deg}^\eta(G_2) = ak^{1/3}.$$

The center quotient defining B_{np}^η equals a , while every leaf quotient equals $ak^{1/2}$. Therefore

$$B_{\text{np}}^\eta(G_2) = ak^{1/2} > ak^{1/3} = U_{\deg}^\eta(G_2).$$

Both examples are connected, which proves the claimed incomparability. $\square$

Proposition 4.10 (Incomparability with the average 2-degree bound). *Let H be a connected r -uniform hypergraph and put*

$$a = \frac{\eta_r}{r}.$$

For a vertex i , let

$$m_i = \frac{\sum_{e \ni i} \prod_{j \in e \setminus \{i\}} d_j}{d_i^{r-1}}$$

be its average 2-degree, and write $m_{(1)} \geq m_{(2)} \geq \dots$. Under the present normalization, the Lin–Mo–Zhou–Weng bound [13] becomes

$$\rho(\mathcal{A}_\eta(H)) \leq U_{\text{av}2}^\eta(H) := a m_{(1)}^{1/r} m_{(2)}^{1-1/r}.$$

Neither $B_{\text{np}}^\eta(H) \leq U_{\text{av}2}^\eta(H)$ nor $U_{\text{av}2}^\eta(H) \leq B_{\text{np}}^\eta(H)$ holds for all connected uniform hypergraphs. Again, incomparability already occurs for $r = 3$.

Proof. For one direction, use the hypergraph G_1 from Proposition 4.9. Its degrees are $(3, 3, 2, 2, 2)$, and direct substitution gives average 2-degrees

$$2, \quad 2, \quad \frac{13}{4}, \quad \frac{13}{4}, \quad \frac{13}{4}.$$

Hence

$$U_{\text{av}2}^\eta(G_1) = \frac{13a}{4}, \quad B_{\text{np}}^\eta(G_1) = \frac{5a}{2},$$

so

$$B_{\text{np}}^\eta(G_1) < U_{\text{av}2}^\eta(G_1).$$

For the reverse direction, let G_3 be the 3-uniform hypergraph on $\{1, \dots, 7\}$ with

$$E(G_3) = \{\{1, 2, 7\}, \{2, 3, 4\}, \{1, 3, 7\}, \{2, 4, 7\}, \{1, 4, 7\}, \{3, 5, 6\}, \{1, 2, 4\}\}.$$

This hypergraph is connected and has degree vector

$$(4, 4, 3, 4, 1, 1, 4).$$

Its average 2-degrees are

$$\frac{15}{4}, \quad \frac{15}{4}, \quad \frac{11}{3}, \quad \frac{15}{4}, \quad 3, \quad 3, \quad \frac{15}{4}.$$

Thus $m_{(1)} = m_{(2)} = 15/4$, and therefore

$$U_{\text{av}2}^\eta(G_3) = \frac{15a}{4}.$$

The neighbor-profile quotients, divided by a , are

$$3 + \frac{\sqrt{3}}{2}, \quad 3 + \frac{\sqrt{3}}{2}, \quad 3, \quad 3 + \frac{\sqrt{3}}{2}, \quad \sqrt{3}, \quad \sqrt{3}, \quad 3 + \frac{\sqrt{3}}{2}.$$

Consequently

$$B_{\text{np}}^\eta(G_3) = a \left(3 + \frac{\sqrt{3}}{2} \right) > \frac{15a}{4} = U_{\text{av}2}^\eta(G_3),$$

because $\sqrt{3} > 3/2$.

The two connected examples prove the claimed incomparability. $\square$

Remark 4.11 (Two distinct classical comparisons). Propositions 4.9 and 4.10 show that the edgewise geometric aggregation in B_{np}^η is genuinely different from both the two-largest-degree statistic and the classical average-2-degree statistic. For each of those two classical estimates, connected 3-uniform examples occur in both comparison directions: the classical estimate can be sharper than B_{np}^η , and B_{np}^η can be sharper than it. These statements concern the indicated classical bounds only; no universal ordering with later degree/co-degree refinements is claimed. The loose-star calculation in Section 7 instead makes the internal comparison with the profile row-sum baseline quantitative.

5. EQUALITY, STRICTNESS, AND QUANTITATIVE DEFECT

Section 4 established that edge-local placement contains spectral information beyond vertex profiles and is not reducible to standard uniform degree statistics. We now ask when the recovery supplied by B_{np}^η becomes ineffective. This leads first to an exact characterization of equality with the profile row-sum bound and then to quantitative estimates of the loss away from equality.

5.1. Exact equality and strictness.

Definition 5.1 (Maximum row-sum set). Let

$$M = \max_i R_i^\eta.$$

The maximum row-sum set of H is

$$V_M = \{i \in V : R_i^\eta = M\}.$$

Proposition 5.2 (Equality criterion on the maximum row-sum set). *Let H have no isolated vertices and let*

$$M = \max_i R_i^\eta.$$

Then

$$B_{\text{np}}^\eta(H) = M$$

if and only if there exists a vertex $i \in V$ such that every incident edge $e \ni i$ satisfies

$$\begin{cases} e \setminus \{i\} \subseteq V_M, & |e| = r, \\ e \subseteq V_M, & |e| < r. \end{cases}$$

Equivalently, one of the Collatz–Wielandt quotients defining $B_{\text{np}}^\eta(H)$ suffers no monomial-level loss in the proof of

$$B_{\text{np}}^\eta(H) \leq M.$$

Proof. For each vertex i , write Q_i for the Collatz–Wielandt quotient appearing in (4.3), so that

$$B_{\text{np}}^\eta(H) = \max_i Q_i.$$

In the proof of Theorem 4.5, every valid monomial

$$z_{i_2} \cdots z_{i_r}$$

is bounded above by M . Since every coefficient in Q_i is positive, we have $Q_i = M$ if and only if every valid monomial appearing in every incident-edge polynomial $P_{r,e,i}$ equals M . Indeed, if all these monomials equal M , then the contribution of an incident edge e of size s is

$$\frac{\eta_s}{s!S(r,s)} M N_{r,s} = M \frac{\eta_s}{s},$$

and summing over $e \ni i$ gives the numerator $M R_i^\eta$, hence $Q_i = M$. Conversely, if one valid monomial is strictly smaller than M , positivity of all coefficients forces $Q_i < M$.

Because $z_j^{r-1} = R_j^\eta \leq M$, every factor satisfies $z_j \leq M^{1/(r-1)}$. A product of exactly $r-1$ such positive factors therefore equals M if and only if every factor equals $M^{1/(r-1)}$, equivalently, every tail vertex occurring in that monomial lies in V_M .

Now fix an edge $e \ni i$. If $|e| = r$, then every valid tail tuple is a permutation of $e \setminus \{i\}$, so all monomials generated by e equal M if and only if

$$e \setminus \{i\} \subseteq V_M.$$

If $|e| < r$, every vertex of $e \setminus \{i\}$ occurs in every valid tail tuple. In addition, there is at least one valid tail tuple containing i in the tail: place every vertex of $e \setminus \{i\}$ once and fill one of the remaining $r - |e| \geq 1$ positions with i (the other remaining positions, if any, may be filled arbitrarily from e). Hence all monomials generated by e equal M if and only if

$$e \subseteq V_M.$$

Thus some Q_i equals M exactly under the stated incident-edge condition. Since every $Q_i \leq M$, this is equivalent to $B_{\text{np}}^\eta(H) = M$. $\square$

Remark 5.3 (A genuinely nonuniform equality feature). In the uniform case $|e| = r$, the first vertex i does not appear in any valid tail tuple for $P_{r,e,i}$; each tail tuple is a permutation of $e \setminus \{i\}$. For lower-order edges $|e| < r$, repetitions are forced, and valid tail tuples may include i itself. Thus the equality criterion in Proposition 5.2 reflects a genuinely nonuniform feature of the blowup model.

Remark 5.4 (A quick strictness test). The proof of Theorem 4.5 gives a useful sufficient condition without introducing an additional bound. If, for every vertex i , some incident edge $e \ni i$ contains a vertex $j \neq i$ with $R_j^\eta < M := \max_v R_v^\eta$, then at least one monomial in the corresponding edge polynomial is strictly smaller than M , and hence $B_{\text{np}}^\eta(H) < M$. This gives a quick sufficient test for strict improvement.

Remark 5.5 (Row-regular case). If

$$R_1^\eta = \dots = R_n^\eta = R,$$

then

$$\mathcal{A}_\eta(H) \mathbf{1}^{r-1} = R \mathbf{1}^{[r-1]},$$

so $\rho(\mathcal{A}_\eta(H)) = R$; when $R > 0$, substitution in (4.3) also gives $B_{\text{np}}^\eta(H) = R$. Conversely, suppose $\mathcal{A}_\eta(H)$ is weakly irreducible and $\rho(\mathcal{A}_\eta(H)) = M := \max_i R_i^\eta$. Let $x > 0$ be a Perron vector and choose p with $x_p = \max_i x_i$. Then

$$\rho x_p^{r-1} = (\mathcal{A}_\eta(H) x^{r-1})_p \leq R_p^\eta x_p^{r-1} \leq M x_p^{r-1} = \rho x_p^{r-1}.$$

Hence equality holds throughout. In particular, every tail coordinate occurring in a positive entry of the p th slice has value x_p . Repeating this argument along the strongly connected representation digraph propagates the maximum value to every coordinate, so x is constant. The eigen-equations then give $R_i^\eta = \rho$ for every i . Thus, under weak irreducibility, equality in the profile row-sum bound is equivalent to η -row-regularity, whereas equality in the neighbor-profile bound is governed by Proposition 5.2.

5.2. Quantitative defect. When the equality condition fails, the next estimates quantify a computable part of the resulting loss. The first bound is obtained monomial by monomial from the arithmetic–geometric mean inequality; the second removes the need to enumerate valid tail words.

Theorem 5.6 (Quantitative neighbor-defect bound). *Let H have no isolated vertices, and set*

$$M = \max_i R_i^\eta, \quad \delta_i = M - R_i^\eta.$$

For each vertex i , define

$$(5.1) \quad D_i^\eta = \frac{1}{(r-1)R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} \sum_{(i_2, \dots, i_r) \in \mathcal{T}(e, i)} \sum_{m=2}^r \delta_{i_m}.$$

Then

$$(5.2) \quad B_{\text{np}}^\eta(H) \leq M - \min_{i \in V} D_i^\eta.$$

In particular, if $D_i^\eta > 0$ for every i , then

$$B_{\text{np}}^\eta(H) < M.$$

Proof. Let

$$z_j = (R_j^\eta)^{1/(r-1)}.$$

For any valid tail tuple $(i_2, \dots, i_r) \in \mathcal{T}(e, i)$, we have

$$z_{i_2} \cdots z_{i_r} = M \prod_{m=2}^r \left(\frac{R_{i_m}^\eta}{M} \right)^{1/(r-1)}.$$

By the arithmetic–geometric mean inequality,

$$\prod_{m=2}^r \left(\frac{R_{i_m}^\eta}{M} \right)^{1/(r-1)} \leq \frac{1}{r-1} \sum_{m=2}^r \frac{R_{i_m}^\eta}{M}.$$

Therefore

$$z_{i_2} \cdots z_{i_r} \leq \frac{1}{r-1} \sum_{m=2}^r R_{i_m}^\eta = M - \frac{1}{r-1} \sum_{m=2}^r \delta_{i_m}.$$

Substituting this estimate into the i th Collatz–Wielandt quotient gives

$$\begin{aligned} & \frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} P_{r,e,i}(z) \\ & \leq \frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} \sum_{(i_2, \dots, i_r) \in \mathcal{T}(e, i)} \left(M - \frac{1}{r-1} \sum_{m=2}^r \delta_{i_m} \right). \end{aligned}$$

The contribution of the M -term is

$$\frac{1}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} M N_{r,|e|} = \frac{M}{R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|} = M.$$

The remaining term is exactly D_i^η . Thus the i th quotient is at most $M - D_i^\eta$. Taking the maximum over i gives

$$B_{\text{np}}^\eta(H) \leq \max_i (M - D_i^\eta) = M - \min_i D_i^\eta.$$

The strictness statement follows immediately. $\square$

Remark 5.7. The quantity D_i^η is a computable linear loss obtained from the arithmetic–geometric mean estimate applied monomial by monomial; it is not claimed to equal the exact loss in the i th Collatz–Wielandt quotient.

Theorem 5.8 (Closed-form local defect formula). *Let H have no isolated vertices, and set*

$$M = \max_i R_i^\eta, \quad \delta_i = M - R_i^\eta.$$

For $2 \leq s \leq r$, define

$$A_{r,s}^{\text{self}} = (r-1) \sum_{t=0}^{s-1} (-1)^t \binom{s-1}{t} (s-t)^{r-2},$$

and

$$A_{r,s}^{\text{out}} = (r-1) \sum_{t=0}^{s-2} (-1)^t \binom{s-2}{t} (s-t)^{r-2}.$$

Then the defect term in Theorem 5.6 admits the closed form

$$D_i^\eta = \frac{1}{(r-1)R_i^\eta} \sum_{e \ni i} \frac{\eta_{|e|}}{|e|! S(r, |e|)} \left(A_{r,|e|}^{\text{self}} \delta_i + A_{r,|e|}^{\text{out}} \sum_{j \in e \setminus \{i\}} \delta_j \right).$$

Consequently, the defect estimate

$$B_{\text{np}}^\eta(H) \leq M - \min_i D_i^\eta$$

can be evaluated without enumerating valid tail words.

Proof. Fix $e \ni i$ with $|e| = s$. Since all edges in this paper have size at least two, $s \geq 2$. We count the total number of appearances of each vertex in the tail positions over all words in $\mathcal{T}(e, i)$.

First consider the distinguished vertex i . Fix one tail position and force it to equal i . The remaining $r-2$ positions must contain every vertex of $e \setminus \{i\}$ at least once. By inclusion-exclusion, the number of such words is

$$\sum_{t=0}^{s-1} (-1)^t \binom{s-1}{t} (s-t)^{r-2}.$$

Since there are $r-1$ tail positions, the total number of appearances of i in all tail positions over all valid words is

$$A_{r,s}^{\text{self}}.$$

Now fix $j \in e \setminus \{i\}$. If one tail position is forced to be j , then the remaining $r-2$ positions must contain every vertex of $e \setminus \{i, j\}$ at least once, while i and j may appear freely. Hence the number of such words for a fixed tail position is

$$\sum_{t=0}^{s-2} (-1)^t \binom{s-2}{t} (s-t)^{r-2}.$$

Multiplying by $r-1$ gives the total number of appearances of the fixed vertex j in all tail positions over all valid words:

$$A_{r,s}^{\text{out}}.$$

Therefore

$$\sum_{(i_2, \dots, i_r) \in \mathcal{T}(e, i)} \sum_{m=2}^r \delta_{i_m} = A_{r,s}^{\text{self}} \delta_i + A_{r,s}^{\text{out}} \sum_{j \in e \setminus \{i\}} \delta_j.$$

Substituting this identity into (5.1) gives the claimed closed form. As a consistency check, counting all $r-1$ tail positions over all $N_{r,s}$ valid words also gives

$$A_{r,s}^{\text{self}} + (s-1)A_{r,s}^{\text{out}} = (r-1)N_{r,s}.$$

□

Remark 5.9 (Uniform specialization of the closed-form defect). When $s = r$, the valid tail words contain no occurrence of the distinguished vertex i . Correspondingly,

$$A_{r,r}^{\text{self}} = 0.$$

Moreover,

$$A_{r,r}^{\text{out}} = (r-1)!,$$

because each vertex in $e \setminus \{i\}$ appears exactly once in each of the $(r-1)!$ valid tail permutations. Thus Theorem 5.8 reduces in the uniform case to the expected defect contribution from the other $r-1$ vertices of each edge.

6. EDGE-LOCAL ORGANIZATION AS EXACT QUOTIENT MACHINERY

The edge-local organization used above has a second role: it supplies the exact reduction machinery needed for the structured families in Section 7. Tensor equitable partitions already provide the general lifting mechanism [11]; related hypergraph notions include perfect colourings and quotient constructions [25, 2]. We therefore keep the general framework brief and focus on the model-specific point: a directly checkable edge-type condition on a mixed-size hypergraph that guarantees tensor equitability for the present support-exact blowup tensor.

6.1. Tensor-equitable partitions. Let $\mathcal{A} = (a_{i_1 \dots i_r})$ be a nonnegative tensor of order r and dimension n . Let

$$\Pi = \{C_1, \dots, C_m\}$$

be a partition of $V = \{1, \dots, n\}$.

Definition 6.1 (Tensor-equitable partition). The partition Π is called tensor-equitable for $\mathcal{A}$ if, for every $a \in \{1, \dots, m\}$, every $i \in C_a$, and every $(a_2, \dots, a_r) \in \{1, \dots, m\}^{r-1}$, the quantity

$$q_{a,a_2,\dots,a_r} = \sum_{i_2 \in C_{a_2}} \cdots \sum_{i_r \in C_{a_r}} a_{ii_2 \dots i_r}$$

is independent of the choice of $i \in C_a$.

When Π is tensor-equitable, define

$$\mathcal{A}/\Pi = (q_{a,a_2,\dots,a_r})_{1 \leq a,a_2,\dots,a_r \leq m}.$$

If $y \in \mathbb{R}^m$ is lifted by $x_i = y_a$ for $i \in C_a$, then

$$(\mathcal{A}x^{r-1})_i = ((\mathcal{A}/\Pi)y^{r-1})_a, \quad i \in C_a.$$

Thus the quotient preserves the tensor action on cell-constant vectors; it need not be symmetric even when $\mathcal{A}$ is symmetric. We use the standard equitable-partition consequence of Jin–Zhang–Zhang [11, Theorem 3.3, Corollary 3.4, Lemma 4.1].

Lemma 6.2 (Equitable-quotient reduction). *Let $\mathcal{A}$ be a weakly irreducible nonnegative tensor of order r and dimension n . Let*

$$\Pi = \{C_1, \dots, C_m\}$$

be a tensor-equitable partition of $[n]$ for $\mathcal{A}$. Then

$$\rho(\mathcal{A}) = \rho(\mathcal{A}/\Pi).$$

Moreover, if $y \in \mathbb{R}_{++}^m$ is a positive Perron vector of $\mathcal{A}/\Pi$, then the lifted vector

$$x_i = y_a, \quad i \in C_a,$$

is a positive Perron vector of $\mathcal{A}$.

Proof. By the equitable-partition results of Jin–Zhang–Zhang [11], quotient eigenpairs lift to cell-constant eigenpairs, and weak irreducibility of $\mathcal{A}$ implies weak irreducibility of the quotient $\mathcal{A}/\Pi$. Hence $\mathcal{A}/\Pi$ has a positive Perron vector y . Its lift $x > 0$ is an eigenvector of $\mathcal{A}$ with eigenvalue $\rho(\mathcal{A}/\Pi)$. Applying the Collatz–Wielandt inequalities (2.2)–(2.3) to this positive vector gives

$$\rho(\mathcal{A}) = \rho(\mathcal{A}/\Pi).$$

The same lifting gives the asserted Perron vector of $\mathcal{A}$. $\square$

6.2. Edge-type-equitable partitions for edge-total-mass tensors. We now specialize the preceding quotient reduction to the tensor $\mathcal{A}_\eta(H)$. The following condition is stated directly in terms of the hypergraph.

Let

$$\Pi = \{C_1, \dots, C_m\}$$

be a partition of V . For an edge $e \in E$, define its cell-count vector by

$$\tau_\Pi(e) = (|e \cap C_1|, \dots, |e \cap C_m|).$$

Thus

$$|\tau_\Pi(e)|_1 = |e|.$$

Definition 6.3 (Edge-type-equitable partition). A partition $\Pi = \{C_1, \dots, C_m\}$ is called edge-type-equitable for H if, for every cell C_a , every vertex $i \in C_a$, every edge size $s \in \mathcal{S}(H)$, and every cell-count vector $\tau \in \mathbb{Z}_{\geq 0}^m$ with $|\tau|_1 = s$, the number

$$N_{a,s,\tau}(i) = |\{e \in E_s : i \in e, \tau_\Pi(e) = \tau\}|$$

is independent of the choice of $i \in C_a$.

Lemma 6.4 (Cell-type tail count). *Fix a partition $\Pi = \{C_1, \dots, C_m\}$. Let $i \in C_a$, let $e \ni i$ have cell-count vector $\tau = \tau_\Pi(e)$, and fix a tail cell sequence $(a_2, \dots, a_r)$. Set*

$$m_b = |\{\ell \in \{2, \dots, r\} : a_\ell = b\}|.$$

The number of tail words

$$(i_2, \dots, i_r) \in C_{a_2} \times \dots \times C_{a_r}$$

such that $\text{supp}(i, i_2, \dots, i_r) = e$ depends only on $r, a, (a_2, \dots, a_r)$, and τ , not on the particular edge e or the particular vertex $i \in C_a$. More explicitly, this number is

$$K_{a, (a_2, \dots, a_r), \tau}^{(r)} = K_a \prod_{b \neq a} K_b,$$

where

$$K_a = \sum_{h=0}^{\tau_a-1} (-1)^h \binom{\tau_a-1}{h} (\tau_a - h)^{m_a}$$

and, for $b \neq a$,

$$K_b = \begin{cases} \tau_b! S(m_b, \tau_b), & m_b \geq \tau_b, \\ 0, & m_b < \tau_b. \end{cases}$$

The convention is that $0!S(0, 0) = 1$, so if $\tau_b = m_b = 0$ then $K_b = 1$. For the distinguished cell, the displayed inclusion–exclusion formula also automatically gives $K_a = 0$ when there are too few tail positions to cover the $\tau_a - 1$ required vertices.

Proof. For each cell C_b , the tail positions with cell label b can only be filled by vertices of $e \cap C_b$, and choices from different cells are independent. If $b \neq a$, all τ_b vertices of $e \cap C_b$ must appear at least once in the m_b positions with label b . This gives the number of surjections from an m_b -set onto a τ_b -set, namely $\tau_b! S(m_b, \tau_b)$, with value zero when $m_b < \tau_b$.

For the distinguished cell C_a , the vertex i is already present as the first coordinate. The tail positions labelled a must contain every vertex of $(e \cap C_a) \setminus \{i\}$, while the distinguished vertex i itself may appear any number of times, including zero. Inclusion–exclusion over the $\tau_a - 1$ non-distinguished vertices gives

$$K_a = \sum_{h=0}^{\tau_a-1} (-1)^h \binom{\tau_a-1}{h} (\tau_a - h)^{m_a}.$$

Multiplying the independent cell-wise counts proves the formula and the claimed independence. In particular, the count depends only on the cell multiplicities and not on the identities of the vertices inside the cells. $\square$

Proposition 6.5 (Edge-type equitability implies tensor equitability). *Let H be a finite simple hypergraph and let $\mathcal{A}_\eta(H)$ be its edge-total-mass normalized tensor. If Π is edge-type-equitable for H , then Π is tensor-equitable for $\mathcal{A}_\eta(H)$.*

Proof. Fix a cell C_a , a vertex $i \in C_a$, and a tail cell sequence

$$(a_2, \dots, a_r) \in \{1, \dots, m\}^{r-1}.$$

We must show that

$$\sum_{i_2 \in C_{a_2}} \dots \sum_{i_r \in C_{a_r}} (\mathcal{A}_\eta(H))_{ii_2 \dots i_r}$$

is independent of the choice of $i \in C_a$.

By the support-exact definition of $\mathcal{A}_\eta(H)$, every nonzero term in this sum is generated by the unique edge

$$e = \text{supp}(i, i_2, \dots, i_r) \in E(H).$$

Thus grouping terms by their support edge neither omits nor double-counts any nonzero tensor entry. If $|e| = s$, then the corresponding tensor weight is

$$\frac{\eta_s}{s!S(r, s)}.$$

Group the incident edges $e \ni i$ according to their size s and their cell-count vector $\tau = \tau_\Pi(e)$. By Lemma 6.4, for fixed data

$$r, \quad a, \quad (a_2, \dots, a_r), \quad \tau,$$

the number of tail words

$$(i_2, \dots, i_r) \in C_{a_2} \times \dots \times C_{a_r}$$

satisfying

$$\text{supp}(i, i_2, \dots, i_r) = e$$

is the purely combinatorial number

$$K_{a, (a_2, \dots, a_r), \tau}^{(r)},$$

which depends only on these data and not on the particular vertex $i \in C_a$ or the particular edge e with cell-count vector τ .

The quotient entry associated with the first cell C_a and tail cell sequence $(a_2, \dots, a_r)$ is therefore

$$\sum_{s \in \mathcal{S}(H)} \sum_{\substack{\tau \in \mathbb{Z}_{\geq 0}^m \\ |\tau|_1 = s, \tau_a \geq 1}} N_{a, s, \tau}(i) \frac{\eta_s}{s! S(r, s)} K_{a, (a_2, \dots, a_r), \tau}^{(r)}.$$

By edge-type equitability, $N_{a, s, \tau}(i)$ is independent of the choice of $i \in C_a$. All other factors in the displayed expression are independent of i . Hence the quotient entry is independent of i , and Π is tensor-equitable for $\mathcal{A}_\eta(H)$. $\square$

Corollary 6.6 (Exact reduction for edge-type-equitable hypergraphs). *Let H have connected 2-section, and let Π be edge-type-equitable for H . Then*

$$\rho(\mathcal{A}_\eta(H)) = \rho(\mathcal{A}_\eta(H)/\Pi).$$

Proof. If the 2-section of H is connected, then $\mathcal{A}_\eta(H)$ is weakly irreducible. By Proposition 6.5, Π is tensor-equitable for $\mathcal{A}_\eta(H)$. The claim follows from Lemma 6.2. $\square$

Remark 6.7 (Strength and scope of edge-type equitability). Definition 6.3 is a directly checkable sufficient hypergraph-level condition; no converse to Proposition 6.5 is asserted. Tensor equitability constrains contributions only after aggregation over a tail-cell sequence, whereas edge-type equitability resolves them first by edge size and cell-count vector, so it is stronger and independent of the particular positive weights η_s . It is also stronger than equality of vertex edge-size profiles, which records incident edge counts but not their distribution among partition cells.

7. SIZE-DEPENDENT SPECTRAL SCALING IN LOOSE STARS

7.1. Multi-size loose stars and competing-size scaling. The same support-exact encoding that makes edge-local arrangement spectrally visible also forces repeated indices for lower-order edges; the latter mechanism changes the asymptotic spectral scale. In a fixed order- r blowup tensor, an s -edge with $s < r$ contributes repeated-vertex monomials, and these repetitions change the spectral exponent of loose-star families. Spectral extremal problems for nearly uniform supertrees under general-hypergraph adjacency tensors have already been studied [28]. Our purpose is different: we make no extremal claim, but fix the loose-star geometry, allow multiple active edge sizes with size-homogeneous masses, and determine the natural competing scales, including regimes in which several edge sizes contribute at leading order. The geometry is also different from the hyperflowers considered by Galuppi–Mulas–Venturello [9]: in the uniform case our edges share one center and otherwise have disjoint leaves, rather than sharing $r - 1$ central vertices. This gives a fully computable family in which lower-order edge sizes are genuine spectral data and cannot be treated as small perturbations of the uniform part.

Definition 7.1 (Rank- r multi-size loose star). Fix $r \geq 3$ and a nonempty set

$$S \subseteq \{2, \dots, r\}$$

with $r \in S$. For each $s \in S$, let $k_s \geq 1$. The rank- r multi-size loose star

$$L_{\mathbf{k}}^{S,r}$$

has one center vertex c . For each $s \in S$, it has k_s pendant s -edges, each consisting of c and $s - 1$ new leaf vertices. All leaf vertices are distinct across all edges.

For $s \in S$, set

$$w_s = \frac{\eta_s}{s!S(r, s)}.$$

Define

$$C_{r,s}(t) = w_s \sum_{\substack{m_0 \geq 0, m_1, \dots, m_{s-1} \geq 1 \\ m_0 + m_1 + \dots + m_{s-1} = r-1}} \frac{(r-1)!}{m_0!m_1! \dots m_{s-1}!} t^{m_1 + \dots + m_{s-1}},$$

and

$$L_{r,s}(t) = w_s \sum_{\substack{m_0 \geq 1, m_1 \geq 0, m_2, \dots, m_{s-1} \geq 1 \\ m_0 + m_1 + \dots + m_{s-1} = r-1}} \frac{(r-1)!}{m_0!m_1! \dots m_{s-1}!} t^{m_1 + \dots + m_{s-1}}.$$

The polynomial $C_{r,s}$ is the contribution of one pendant s -edge to the center equation after normalizing the center value to one. The polynomial $L_{r,s}$ is the corresponding contribution to the equation of a distinguished leaf on that s -edge.

Theorem 7.2 (Exact Perron system for multi-size loose stars). *Let $L_{\mathbf{k}}^{S,r}$ be a rank- r multi-size loose star. For every $s \in S$ with $k_s > 0$, let t_s denote the ratio between a leaf value on an s -edge and the center value in the positive Perron vector. Then $\lambda = \rho(\mathcal{A}_\eta(L_{\mathbf{k}}^{S,r}))$ satisfies*

$$\lambda = \sum_{s \in S} k_s C_{r,s}(t_s),$$

and

$$\lambda t_s^{r-1} = L_{r,s}(t_s), \quad s \in S, \quad k_s > 0.$$

Conversely, any positive solution of this system gives a positive eigenpair of the quotient tensor associated with the natural partition into the center and the leaf types. Since this quotient tensor is weakly irreducible on the active quotient classes, the corresponding eigenvalue is its spectral radius, and the exact quotient reduction lifts it to the spectral radius of the full tensor.

Proof. The natural partition has one singleton cell for the center and, for each active size s , one cell containing all leaves belonging to s -edges. It is edge-type-equitable: every s -leaf lies in exactly one incident edge, and that edge has one center vertex and $s - 1$ vertices in the s -leaf cell; the center cell is a singleton. The 2-section of the multi-size loose star is connected, so $\mathcal{A}_\eta(L_{\mathbf{k}}^{S,r})$ is weakly irreducible. Proposition 6.5 and Lemma 6.2 therefore give a positive quotient Perron vector whose lift is the positive Perron vector of the full tensor. In particular, the Perron vector is constant on each leaf-type cell.

Normalize the center value to one. For a pendant s -edge, write its vertices as

$$\{c, u_1, \dots, u_{s-1}\},$$

with all leaf values equal to t_s . In the center equation, a valid tail word of length $r - 1$ contains each leaf at least once and may contain the center any number of times. Grouping such words by the multiplicities $(m_0, m_1, \dots, m_{s-1})$ gives $C_{r,s}(t_s)$. Summing over the k_s edges of size s gives the center equation.

For the equation of a distinguished leaf, say u_1 , the full word already contains u_1 in the first coordinate. A valid tail must contain the center at least once and must contain every other leaf at least once, while u_1 may appear any number of times in the tail. Grouping tail words by multiplicities gives $L_{r,s}(t_s)$. Since the left side of the leaf eigen-equation is λt_s^{r-1} , the leaf equations follow.

The converse is obtained by assigning the center value one and all leaves on s -edges the value t_s . The preceding calculation verifies the quotient tensor eigen-equations. The active quotient tensor is weakly irreducible because each leaf type is connected to the center type and the

center type is connected back to every active leaf type. Hence a positive quotient eigenpair has eigenvalue equal to the quotient spectral radius by the Collatz–Wielandt theorem. The exact quotient reduction then lifts this spectral radius to the full tensor. $\square$

For the asymptotic statement, define

$$q_s = r - s + 1, \\ a_{r,s} = w_s \frac{(r-1)!}{(r-s)!}, \quad b_{r,s} = w_s \frac{(r-1)!}{(r-s+1)!},$$

and

$$K_{r,s} = a_{r,s} b_{r,s}^{(s-1)/q_s}.$$

Lemma 7.3 (Balancing scales). *Let S be finite. For each $s \in S$, let $q_s > 0$, $K_s > 0$, and $k_s > 0$, and put*

$$d_s = \frac{q_s + s - 1}{q_s}.$$

Suppose $\lambda \rightarrow \infty$ and

$$\lambda = \sum_{s \in S} K_s k_s \lambda^{-(s-1)/q_s} (1 + \varepsilon_s), \quad \max_{s \in S} |\varepsilon_s| \rightarrow 0.$$

Set

$$M_s = (K_s k_s)^{1/d_s}, \quad M = \max_{s \in S} M_s.$$

Then $\lambda \asymp M$. If, in addition,

$$\frac{M_s}{M} \rightarrow \theta_s \quad (s \in S),$$

then $\max_s \theta_s = 1$ and

$$\frac{\lambda}{M} \rightarrow y_\theta,$$

where y_θ is the unique positive solution of

$$(7.1) \quad 1 = \sum_{s \in S} \theta_s^{d_s} y^{-d_s}.$$

In particular, if $M_s = o(M_j)$ for every $s \neq j$, then $\lambda/M_j \rightarrow 1$.

Proof. For all sufficiently large parameters, the factors $1 + \varepsilon_s$ lie between $1/2$ and 2 . To prove $\lambda = O(M)$, suppose otherwise. Then along some subsequence $\lambda/M \rightarrow \infty$, and for every s ,

$$\frac{K_s k_s \lambda^{-(s-1)/q_s}}{\lambda} = \left(\frac{M_s}{\lambda} \right)^{d_s} \leq \left(\frac{M}{\lambda} \right)^{d_s} \rightarrow 0.$$

After division by λ , the defining equation would have right-hand side tending to zero, a contradiction. Similarly, if $\lambda \not\asymp M$, there is a subsequence with $\lambda/M \rightarrow 0$. Choose, along a further subsequence, a fixed index s_0 with $M_{s_0} = M$, which is possible because S is finite. The s_0 -term divided by λ is then $(M/\lambda)^{d_{s_0}} (1 + \varepsilon_{s_0}) \rightarrow \infty$, again a contradiction. Hence $\lambda \asymp M$.

Now assume $M_s/M \rightarrow \theta_s$ for every s . For each parameter value some index attains $M_s = M$; by finiteness of S , one such index occurs along an infinite subsequence. The assumed convergence of all ratios then implies that at least one $\theta_s = 1$, while every $\theta_s \leq 1$. Thus $\max_s \theta_s = 1$. Put $y = \lambda/M$. The first part keeps y in a compact subinterval of $(0, \infty)$. Dividing the defining equation by λ gives

$$1 = \sum_{s \in S} \left(\frac{M_s}{M} \right)^{d_s} y^{-d_s} (1 + \varepsilon_s).$$

Every subsequential limit $y_0 > 0$ therefore satisfies

$$1 = \sum_{s \in S} \theta_s^{d_s} y_0^{-d_s}.$$

The right-hand side is a continuous strictly decreasing function of $y_0 > 0$, tending to infinity as $y_0 \downarrow 0$ and to zero as $y_0 \rightarrow \infty$. Thus (7.1) has a unique positive solution y_θ , so every subsequence

has the same limit and $\lambda/M \rightarrow y_\theta$. If $M_s = o(M_j)$ for $s \neq j$, then $\theta_j = 1$ and all other $\theta_s = 0$; the unique solution is $y_\theta = 1$. $\square$

Theorem 7.4 (Scaling law for multi-size loose stars). *Let r , S , and η be fixed, and let $\mathbf{k} = (k_s)_{s \in S}$ range over positive integer vectors. Set*

$$\lambda_{\mathbf{k}} = \rho(\mathcal{A}_\eta(L_{\mathbf{k}}^{S,r})), \quad q_s = r - s + 1,$$

and

$$M_s = (K_{r,s} k_s)^{q_s/r}, \quad M = \max_{s \in S} M_s.$$

Assume that $\max_{s \in S} k_s \rightarrow \infty$. Then

$$\lambda_{\mathbf{k}} \asymp M.$$

If the relative scales converge,

$$\frac{M_s}{M} \rightarrow \theta_s \quad (s \in S),$$

then

$$\frac{\lambda_{\mathbf{k}}}{M} \rightarrow y_\theta,$$

where y_θ is the unique positive solution of

$$(7.2) \quad 1 = \sum_{s \in S} \theta_s^{r/q_s} y^{-r/q_s}.$$

In particular, if for some $j \in S$,

$$k_j \rightarrow \infty, \quad k_s = o(k_j^{q_j/q_s}) \quad (s \neq j),$$

then

$$\lambda_{\mathbf{k}} \sim K_{r,j}^{q_j/r} k_j^{q_j/r}.$$

Thus a dominant family of pendant j -edges has spectral exponent

$$\frac{q_j}{r} = \frac{r - j + 1}{r}.$$

Proof. We first record the lowest-degree terms of the two edge polynomials. In $L_{r,s}$, the smallest power of t occurs when each of the other $s - 2$ leaves appears once, the distinguished leaf appears zero times, and the center fills the remaining $r - s + 1$ tail positions. Hence

$$L_{r,s}(t) = b_{r,s} t^{s-2} + O(t^{s-1}).$$

Similarly,

$$C_{r,s}(t) = a_{r,s} t^{s-1} + O(t^s).$$

In particular, since $q_s = r - s + 1 > 0$,

$$\frac{L_{r,s}(t)}{t^{r-1}} \sim b_{r,s} t^{-q_s} \rightarrow \infty \quad (t \downarrow 0).$$

We now prove that $\lambda_{\mathbf{k}} \rightarrow \infty$. Suppose instead that along a subsequence $\lambda_{\mathbf{k}} \leq C$. For each fixed $s \in S$, the leaf equation

$$\lambda_{\mathbf{k}} = \frac{L_{r,s}(t_s)}{t_s^{r-1}}$$

and the preceding limit show that $t_s \geq \delta_s > 0$ on this subsequence. For each fixed s , positivity of the coefficients gives $C_{r,s}(t_s) \geq C_{r,s}(\delta_s) > 0$. Because S is finite, set

$$c := \min_{s \in S} C_{r,s}(\delta_s) > 0.$$

Then $C_{r,s}(t_s) \geq c$ for every $s \in S$ on the subsequence. Hence the center equation gives

$$\lambda_{\mathbf{k}} = \sum_{s \in S} k_s C_{r,s}(t_s) \geq c \max_{s \in S} k_s \rightarrow \infty,$$

a contradiction.

From the leaf equation

$$\lambda = \frac{L_{r,s}(t_s)}{t_s^{r-1}},$$

we therefore obtain $t_s \rightarrow 0$ for every fixed $s \in S$. Indeed, for any $\varepsilon > 0$, the function $L_{r,s}(t)/t^{r-1}$ is continuous on $[\varepsilon, \infty)$ and tends to zero as $t \rightarrow \infty$, because $\deg L_{r,s} \leq r - 2$. It is therefore bounded on $[\varepsilon, \infty)$. Since $\lambda \rightarrow \infty$, eventually $t_s < \varepsilon$, and ε is arbitrary.

Using the expansions recorded above in the leaf equation gives

$$t_s \sim \left(\frac{b_{r,s}}{\lambda} \right)^{1/q_s}.$$

Substitution into the center-edge expansion then gives

$$C_{r,s}(t_s) \sim K_{r,s} \lambda^{-(s-1)/q_s}.$$

The center equation becomes

$$\lambda = \sum_{s \in S} K_{r,s} k_s \lambda^{-(s-1)/q_s} (1 + o(1)).$$

Here each relative error tends to zero because $t_s \rightarrow 0$; since S is fixed and finite, the maximum of these finitely many relative errors also tends to zero. Thus the $o(1)$ term may be taken uniformly over S .

Since $q_s + s - 1 = r$, Lemma 7.3 applies with $d_s = r/q_s$, and gives $\lambda_{\mathbf{k}} \asymp M$. If $M_s/M \rightarrow \theta_s$, the same lemma gives (7.2) and the normalized limit. Finally, the condition

$$k_s = o(k_j^{q_j/q_s})$$

for $s \neq j$ is equivalent to $M_s = o(M_j)$, because the constants $K_{r,s}$ are fixed and positive. The last part of the lemma then yields

$$\lambda_{\mathbf{k}} \sim M_j = (K_{r,j} k_j)^{q_j/r},$$

which proves the dominant-size formula and its exponent. $\square$

The exponent in Theorem 7.4 has a direct structural interpretation. For fixed tensor order r , changing the dominant edge size s changes the power of the large parameter itself,

$$\frac{r - s + 1}{r},$$

not merely a multiplicative normalization constant. Lower-order edges force more repeated occurrences of the center in the support-exact blowup words, which changes the balance between the center and leaf equations and produces the size-dependent exponent.

Remark 7.5 (Rank-three exponents). For $r = 3$, dominant pendant 2-edges have exponent $2/3$, while dominant pendant 3-edges have exponent $1/3$. Thus lower-order edges can asymptotically dominate uniform edges in the edge-total-mass order-three blowup model.

Remark 7.6 (A balanced rank-three regime). Take $r = 3$ and $\eta_2 = \eta_3 = 1$. Then

$$K_{3,2} = \frac{1}{3\sqrt{6}}, \quad K_{3,3} = \frac{1}{27}.$$

If $k_2 \rightarrow \infty$ and

$$k_3 \sim \frac{1}{2} k_2^2,$$

then the two natural scales are asymptotically equal:

$$M_2 \sim M_3 \sim M = \left(\frac{k_2}{3\sqrt{6}} \right)^{2/3}.$$

Thus $\theta_2 = \theta_3 = 1$, and (7.2) becomes

$$1 = y^{-3/2} + y^{-3}.$$

Writing $\varphi = (1 + \sqrt{5})/2$, the unique positive solution is

$$y = \varphi^{2/3}.$$

Hence

$$\rho(\mathcal{A}_\eta(L_{\mathbf{k}}^{\{2,3\},3})) \sim \varphi^{2/3} \left(\frac{k_2}{3\sqrt{6}} \right)^{2/3}.$$

This is the critical regime between the two single-size dominance laws.

7.2. Loose stars and unbounded neighbor-profile improvement. The adjacency spectral radius of a uniform hyperstar is classical: under the standard uniform adjacency normalization, a hyperstar with k arms has spectral radius $k^{1/r}$ [12]. The scalar factor $a = \eta_r/r$ below only reflects the present normalization. We include that identity to put all three quantities on the same scale; the point of the proposition is the exact neighbor-profile value and its unbounded separation from the row-sum bound.

Proposition 7.7 (Loose-star formulas and asymptotic separation). *Let $S_{r,k}$ be the r -uniform loose star consisting of $k \geq 2$ edges sharing exactly one common center vertex and otherwise pairwise disjoint leaf vertices. Let*

$$a = \frac{\eta_r}{r}.$$

Then, for the edge-total-mass normalized tensor $\mathcal{A}_\eta(S_{r,k})$,

$$\rho(\mathcal{A}_\eta(S_{r,k})) = ak^{1/r},$$

$$B_{\text{np}}^\eta(S_{r,k}) = ak^{1/(r-1)},$$

and

$$\max_i R_i^\eta = ak.$$

Consequently, for fixed $r \geq 3$,

$$\frac{\max_i R_i^\eta}{B_{\text{np}}^\eta(S_{r,k})} = k^{(r-2)/(r-1)} \rightarrow \infty \quad (k \rightarrow \infty).$$

Thus the neighbor-profile bound can improve the profile row-sum bound by an unbounded factor even within the uniform specialization of the present normalization.

Proof. Let c be the center vertex. In the r -uniform case, every incident edge contributes

$$a = \frac{\eta_r}{r}$$

to the profile row sum of each of its vertices. Hence

$$R_c^\eta = ak, \quad R_\ell^\eta = a$$

for every leaf vertex ℓ . Therefore

$$\max_i R_i^\eta = ak.$$

We first compute the spectral radius. The 2-section of the loose star is connected, and the two-cell partition consisting of the center $\{c\}$ and the set of all leaves is edge-type-equitable. By the exact quotient-reduction principle, the positive quotient Perron vector lifts to a positive Perron vector that is constant on all leaves; write its values as x_c at the center and x_ℓ on every leaf.

For an r -uniform edge, the contribution to the tensor eigen-equation at a vertex is

$$a$$

times the product of the other $r - 1$ vertex values in that edge. Thus

$$\lambda x_c^{r-1} = k a x_\ell^{r-1},$$

and

$$\lambda x_\ell^{r-1} = a x_c x_\ell^{r-2}.$$

Let

$$t = \frac{x_c}{x_\ell}.$$

The second equation gives

$$\lambda = at.$$

Substituting into the first equation gives

$$at^r = ka,$$

and hence

$$t = k^{1/r}.$$

Therefore

$$\rho(\mathcal{A}_\eta(S_{r,k})) = \lambda = ak^{1/r}.$$

It remains to compute B_{np}^η . Since

$$z_c = (ak)^{1/(r-1)}, \quad z_\ell = a^{1/(r-1)},$$

the center quotient is

$$B_c = \frac{1}{ak} \sum_{e \ni c} a \prod_{\ell \in e \setminus \{c\}} z_\ell = \frac{1}{ak} \cdot ka \cdot a = a.$$

For a leaf vertex ℓ , its unique incident edge contains the center and $r - 2$ other leaves. Hence its quotient is

$$B_\ell = \frac{1}{a} \cdot az_c z_\ell^{r-2} = z_c z_\ell^{r-2} = (ak)^{1/(r-1)} a^{(r-2)/(r-1)} = ak^{1/(r-1)}.$$

Since $k \geq 2$, the leaf quotient dominates the center quotient. Therefore

$$B_{\text{np}}^\eta(S_{r,k}) = ak^{1/(r-1)}.$$

The separation statement follows immediately. $\square$

Remark 7.8 (Interpretation of the loose-star separation). For $S_{r,k}$, the spectral radius, neighbor-profile bound, and row-sum bound scale respectively as

$$k^{1/r}, \quad k^{1/(r-1)}, \quad k.$$

Thus, within the row-sum/neighbor-profile hierarchy studied here, the one-step profile scaling improves the row-sum baseline by a power-law factor rather than by a fixed constant. No global dominance over other degree- or co-degree-based bounds is asserted.

8. CONCLUSION

For the support-exact edge-total-mass model, vertex profiles determine all slice row sums but not the spectral radius, even with the complete labelled 2-edge layer fixed; the minimal five-vertex separation persists for every positive pair η_2, η_3 . Edge-local placement then contributes spectral information not reducible to those vertex statistics, while lower-order edges force repeated indices that can change the asymptotic power-law scale. Exact quotient reduction links these two structural uses of the support-exact encoding.

The results are confined to finite simple hypergraphs with edges of size at least two and to positive size-homogeneous weights in the Galuppi–Mulas–Venturello-type support-exact model. Other nonuniform tensor encodings and heterogeneous weights among edges of the same size may behave differently. Natural next questions are to test the same information-loss phenomena in those settings and to identify higher-order edge-local statistics beyond the scalar profile-induced row sums used here.

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Wanzheng Ning conceived the study, developed the main theoretical framework and the principal proofs, and drafted the manuscript. Qianzhi Ao contributed to parts of the proofs, supporting calculations and verification, and reviewed the manuscript. Both authors approved the final manuscript.

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REFERENCES

- [1] S. G. Aksoy, I. Amburg, and S. J. Young, Scalable tensor methods for nonuniform hypergraphs, *SIAM Journal on Mathematics of Data Science* 6 (2024), no. 2, 481–503.
- [2] A. Banerjee and S. Parui, On some building blocks of hypergraphs, *Linear and Multilinear Algebra* 73 (2025), no. 7, 1369–1402.
- [3] A. Banerjee, A. Char, and B. Mondal, Spectra of general hypergraphs, *Linear Algebra and its Applications* 518 (2017), 14–30.
- [4] K. C. Chang, K. Pearson, and T. Zhang, Perron–Frobenius theorem for nonnegative tensors, *Communications in Mathematical Sciences* 6 (2008), no. 2, 507–520.
- [5] G. Contreras-Aso, C. Pérez-Corral, and M. Romance, Uplifting edges in higher-order networks: spectral centralities for non-uniform hypergraphs, *AIMS Mathematics* 9 (2024), no. 11, 32045–32075.
- [6] J. Cooper and A. Dutle, Spectra of uniform hypergraphs, *Linear Algebra and its Applications* 436 (2012), 3268–3292.
- [7] C. Duan and L. Wang, Bounds on the spectral radius of general hypergraphs in terms of clique number, *Linear Algebra and its Applications* 610 (2021), 120–134.
- [8] S. Friedland, S. Gaubert, and L. Han, Perron–Frobenius theorem for nonnegative multilinear forms and extensions, *Linear Algebra and its Applications* 438 (2013), 738–749, doi:10.1016/j.laa.2011.02.042.
- [9] F. Galuppi, R. Mulas, and L. Venturello, Spectral theory of weighted hypergraphs via tensors, *Linear and Multilinear Algebra* 71 (2023), no. 3, 317–347, doi:10.1080/03081087.2022.2030659.
- [10] Y. Hou, A. Chang, and L. Zhang, A homogeneous polynomial associated with general hypergraphs and its applications, *Linear Algebra and its Applications* 591 (2020), 72–86.
- [11] Y.-L. Jin, J. Zhang, and X.-D. Zhang, Equitable partition theorem of tensors and spectrum of generalized power hypergraphs, *Linear Algebra and its Applications* 555 (2018), 21–38, doi:10.1016/j.laa.2018.06.002.
- [12] H. Li, J.-Y. Shao, and L. Qi, The extremal spectral radii of k -uniform supertrees, *Journal of Combinatorial Optimization* 32 (2016), no. 3, 741–764.
- [13] H. Lin, B. Mo, B. Zhou, and W. Weng, Sharp bounds for ordinary and signless Laplacian spectral radii of uniform hypergraphs, *Applied Mathematics and Computation* 285 (2016), 217–227, doi:10.1016/j.amc.2016.03.016.
- [14] X. Yuan, M. Zhang, and M. Lu, Some upper bounds on the eigenvalues of uniform hypergraphs, *Linear Algebra and its Applications* 484 (2015), 540–549, doi:10.1016/j.laa.2015.06.023.
- [15] L. Liu, L. Kang, and S. Bai, Bounds on the spectral radius of uniform hypergraphs, *Discrete Applied Mathematics* 259 (2019), 160–169.
- [16] C. Deng, J. Dong, and H. Li, Several bounds on the spectral radius of uniform hypergraphs, *Discrete Applied Mathematics* 382 (2026), 317–326, doi:10.1016/j.dam.2025.12.014.
- [17] H. Li, C. Deng, and C. Bu, On the α -spectral radius of general hypergraphs, *Linear Algebra and its Applications* 752 (2027), 107–130, doi:10.1016/j.laa.2026.08.012.
- [18] C. Li, Y. Li, and X. Kong, Some bounds for the spectral radius of nonnegative tensors, *Journal of Industrial and Management Optimization* 12 (2016), no. 3, 975–983.
- [19] L.-H. Lim, Singular values and eigenvalues of tensors: a variational approach, in *Proceedings of the IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing*, 2005, 129–132.
- [20] M. Ng, L. Qi, and G. Zhou, Finding the largest eigenvalue of a nonnegative tensor, *SIAM Journal on Matrix Analysis and Applications* 31 (2009), no. 3, 1090–1099.
- [21] X. Ouyard, J.-M. Le Goff, and S. Marchand-Maillet, Adjacency and tensor representation in general hypergraphs. Part 1: e-adjacency tensor uniformisation using homogeneous polynomials, arXiv:1712.08189.
- [22] L. Qi, Eigenvalues of a real supersymmetric tensor, *Journal of Symbolic Computation* 40 (2005), 1302–1324.
- [23] L. Qi, Symmetric nonnegative tensors and copositive tensors, *Linear Algebra and its Applications* 439 (2013), no. 1, 228–238, doi:10.1016/j.laa.2013.03.015.
- [24] L. Sun, J. Zhou, and C. Bu, Spectral properties of general hypergraphs, *Linear Algebra and its Applications* 561 (2019), 187–203.
- [25] A. A. Taranenko, Perfect colourings of hypergraphs, *Linear and Multilinear Algebra* 73 (2025), no. 8, 1566–1590.
- [26] Y. Yang and Q. Yang, Further results for Perron–Frobenius theorem for nonnegative tensors, *SIAM Journal on Matrix Analysis and Applications* 31 (2010), no. 5, 2517–2530.
- [27] J. Zhang and C. Bu, An iterative method for finding the spectral radius of an irreducible nonnegative tensor, *Computational and Applied Mathematics* 40 (2021), Paper No. 8, 14 pp.
- [28] W. Zhang, L. Liu, L. Kang, and Y. Bai, Some properties of the spectral radius for general hypergraphs, *Linear Algebra and its Applications* 513 (2017), 103–119.